1. Pollutants emitted directly from identifiable sources into the atmosphere are classified as:

(A) Primary pollutants
(B) Secondary pollutants
(C) Tertiary pollutants
(D) Fugitive complexes

2. Which of the following is an example of a secondary air pollutant formed by atmospheric chemical reactions?

(A) Carbon monoxide (CO)
(B) Sulfur dioxide (SO2)
(C) Ground-level ozone (O3)
(D) Nitric oxide (NO)

3. The poisonous gas that binds irreversibly to blood hemoglobin with 200 times higher affinity than oxygen is:

(A) Carbon dioxide (CO2)
(B) Carbon monoxide (CO)
(C) Sulfur dioxide (SO2)
(D) Methane (CH4)

4. Photochemical smog is primarily formed on warm sunny days by atmospheric reactions between:

(A) SO2 and particulate matter in fog
(B) NOx and Volatile Organic Compounds (VOCs) under UV sunlight
(C) CO2 and water vapor
(D) Methane and ozone

5. The primary secondary pollutant responsible for eye irritation and plant foliar damage in photochemical smog is:

(A) Peroxyacetyl Nitrate (PAN)
(B) Carbon monoxide
(C) Hydrogen sulfide
(D) Ammonium nitrate

6. Classical (London-type) smog is chemically reducing in nature, characterized by high concentrations of:

(A) Ozone and hydrocarbons
(B) Sulfur dioxide (SO2) and carbon soot particulates in cold fog
(C) Nitrogen dioxide
(D) Peroxyacetyl nitrate

7. Normal unpolluted rainwater has a slightly acidic pH of approximately 5.6 due to the dissolution of atmospheric:

(A) Sulfur dioxide
(B) Carbon dioxide forming carbonic acid (H2CO3)
(C) Nitrogen gas
(D) Methane

8. Acid rain is defined as precipitation exhibiting a pH value:

(A) Below 5.6
(B) Above 7.0
(C) Between 6.5 and 7.5
(D) Exactly 8.2

9. The two primary inorganic acids responsible for the acidity of acid rain are:

(A) Hydrochloric acid and acetic acid
(B) Sulfuric acid (H2SO4) and Nitric acid (HNO3)
(C) Phosphoric acid and carbonic acid
(D) Formic acid and citric acid

10. Air particulate matter with aerodynamic diameter of 2.5 microns or smaller is designated as:

(A) PM10
(B) PM2.5
(C) Suspended soot
(D) Coarse dust

11. Which air pollution particulate control device uses high-voltage electrostatic fields (40-70 kV) to charge and collect dust?

(A) Cyclone separator
(B) Electrostatic Precipitator (ESP)
(C) Fabric baghouse filter
(D) Wet venturi scrubber

12. The automotive three-way catalytic converter utilizes which precious metals to catalyze exhaust reduction and oxidation?

(A) Iron and Copper
(B) Platinum (Pt), Palladium (Pd), and Rhodium (Rh)
(C) Aluminum and Titanium
(D) Gold and Silver

13. In a three-way catalytic converter, rhodium primarily catalyzes the reduction of toxic NOx into:

(A) N2 and O2 gases
(B) Ammonia gas
(C) Nitric acid
(D) Nitrous oxide

14. A water pollution source discharged through a distinct, identifiable pipe, outfall ditch, or canal is a:

(A) Non-point source
(B) Point source
(C) Diffuse agricultural runoff
(D) Atmospheric wet deposition

15. Biochemical Oxygen Demand (BOD) measures the amount of dissolved oxygen consumed by microorganisms to decompose:

(A) Inorganic mineral salts
(B) Biodegradable organic matter under aerobic conditions over 5 days at 20 deg C
(C) Heavy metal precipitates
(D) Toxic pesticide complexes

16. Chemical Oxygen Demand (COD) is determined in the laboratory by boiling wastewater with which powerful oxidizing agent?

(A) Potassium permanganate
(B) Potassium dichromate (K2Cr2O7) in concentrated sulfuric acid
(C) Hydrogen peroxide
(D) Sodium hypochlorite

17. For any raw domestic or industrial wastewater, the relationship between COD and BOD is always:

(A) BOD is greater than COD
(B) COD is greater than or equal to BOD
(C) COD is equal to zero
(D) BOD divided by COD is always 10

18. The severe neurological disease caused by eating fish contaminated with methylmercury in Japan is:

(A) Itai-itai disease
(B) Minamata disease
(C) Blackfoot disease
(D) Fluorosis

19. Itai-itai disease, characterized by severe osteomalacia and renal failure in Japan, was caused by water polluted with:

(A) Lead (Pb)
(B) Cadmium (Cd)
(C) Arsenic (As)
(D) Mercury (Hg)

20. Chronic ingestion of groundwater contaminated with excess Arsenic (As) causes which vascular condition?

(A) Skeletal fluorosis
(B) Blackfoot disease
(C) Plumbism
(D) Methemoglobinemia

21. Infant methemoglobinemia ('Blue Baby Syndrome') is caused by drinking groundwater containing excessive concentrations of:

(A) Fluoride
(B) Nitrate (NO3-)
(C) Chloride
(D) Sulfate

22. Skeletal and dental fluorosis in human populations is caused by drinking groundwater with fluoride levels exceeding:

(A) 0.1 mg/L
(B) 1.5 mg/L (WHO permissible limit)
(C) 10.0 mg/L
(D) 50.0 mg/L

23. In primary wastewater treatment, large floating debris (plastics, rags, wood) are mechanically removed by:

(A) Trickling filters
(B) Screening bars and racks
(C) Aeration basins
(D) Chlorine contact tanks

24. The conventional secondary wastewater treatment process utilizing suspended aerobic microbial flocs is the:

(A) Septic tank
(B) Activated Sludge Process (ASP)
(C) Stabilization pond
(D) Deep well injection

25. In the Activated Sludge Process, what is continuously supplied to the aeration basin to sustain aerobic bio-oxidation?

(A) Pure nitrogen gas
(B) Compressed atmospheric air or oxygen
(C) Carbon dioxide
(D) Methane gas

26. Sound pressure level is measured on a logarithmic scale in units of:

(A) Hertz (Hz)
(B) Decibels (dB)
(C) Pascals (Pa)
(D) Watts (W)

27. The threshold of human hearing for a healthy young adult at 1,000 Hz corresponds to a sound level of:

(A) 0 dB
(B) 20 dB
(C) 50 dB
(D) 100 dB

28. Prolonged occupational exposure to continuous sound levels exceeding what threshold causes permanent sensorineural hearing loss?

(A) 55 dBA
(B) 85 dBA (OSHA action level)
(C) 40 dBA
(D) 10 dBA

29. Thermal pollution of natural water bodies is primarily caused by heated cooling water discharged from:

(A) Thermal and nuclear power generation plants
(B) Municipal storm drains
(C) Agricultural irrigation canals
(D) Domestic laundry basins

30. Thermal pollution harms aquatic ecosystems primarily because higher water temperature:

(A) Reduces the saturation solubility of dissolved oxygen (DO) while accelerating fish metabolic rates
(B) Increases water salinity to 100%
(C) Freezes bottom benthic sediments
(D) Destroys all water molecules

31. Which type of nuclear radiation consists of high-energy helium nuclei (2 protons and 2 neutrons) with low penetration power?

(A) Alpha (alpha) particles
(B) Beta (beta) particles
(C) Gamma (gamma) rays
(D) Neutron flux

32. The SI unit of radioactive activity, representing one nuclear disintegration per second, is the:

(A) Curie (Ci)
(B) Becquerel (Bq)
(C) Roentgen (R)
(D) Sievert (Sv)

33. The SI unit of radiation absorbed dose equivalent, quantifying biological damage in human tissue, is the:

(A) Becquerel (Bq)
(B) Gray (Gy)
(C) Sievert (Sv)
(D) Coulomb (C)

34. The worst commercial nuclear power disaster in history, occurring in Soviet Ukraine in April 1986, took place at:

(A) Three Mile Island
(B) Chernobyl Nuclear Power Plant
(C) Fukushima Daiichi
(D) Kashiwazaki

35. The Fukushima Daiichi nuclear accident in March 2011 was initiated by an offshore earthquake followed immediately by a:

(A) Major volcanic eruption
(B) Catastrophic tsunami that disabled emergency diesel generators
(C) Direct aircraft collision
(D) Cyberattack on control software

36. In the waste management hierarchy, what is universally regarded as the most preferred, sustainable option?

(A) Incineration with energy recovery
(B) Source Reduction and Waste Prevention
(C) Recycling
(D) Sanitary landfilling

37. In sanitary landfills, an impermeable bottom liner of compacted clay and synthetic geomembrane is installed to prevent:

(A) Methane gas diffusion into air
(B) Leachate from percolating down into groundwater aquifers
(C) Birds from scavenging garbage
(D) Vegetation growth on the landfill cap

38. The foul, toxic dark liquid generated when rainwater percolates through decomposing municipal solid waste is termed:

(A) Humic effluent
(B) Leachate
(C) Blackwater
(D) Supernatant

39. Under the Indian Bio-Medical Waste Management Rules, human anatomical tissues, organs, and soiled cotton dressings are segregated in:

(A) Red bags
(B) Yellow color-coded bags (destined for incineration)
(C) Blue cardboard boxes
(D) White translucent puncture-proof containers

40. Under Bio-Medical Waste Rules, waste sharps (needles, scalpels, blades) must be collected in:

(A) Yellow bags
(B) Red bags
(C) White translucent, puncture-proof, leak-proof containers
(D) Blue bins

41. The collection efficiency of an Electrostatic Precipitator (ESP) is mathematically governed by the:

(A) Deutsch-Anderson equation: eta = 1 - exp(-w * A / Q)
(B) Arrhenius reaction equation
(C) Darcy-Weisbach friction equation
(D) Ergun packed-bed equation

42. In the Deutsch-Anderson equation, what does the parameter 'w' represent?

(A) Width of the collection plate
(B) Effective migration velocity of the charged particle toward the collecting electrode
(C) Weight of the ash hopper
(D) Wind speed inside the duct

43. In a reverse-pulse jet fabric filter (baghouse), accumulated dust cakes on the outside of fabric bags are dislodged by:

(A) High-velocity water jets
(B) Periodic bursts of high-pressure compressed air pulsing down inside the filter bag
(C) Manual mechanical brushing
(D) Heating bags to 500 deg C

44. A Cyclone separator removes particulate matter from gas streams primarily by exploiting which physical force?

(A) Electrostatic attraction
(B) Centrifugal inertial force impelling particles toward the outer cyclone wall
(C) Magnetic deflection
(D) Ultrasonic cavitation

45. Venturi wet scrubbers achieve high collection efficiency for sub-micron particulates by:

(A) Injecting atomized water droplets into gas accelerated through a converging-diverging venturi throat
(B) Filtering through sand beds
(C) Passing gas over dry limestone blocks
(D) Using activated carbon pellets

46. Flue Gas Desulfurization (FGD) removes sulfur dioxide (SO_2) from coal-fired power plant chimney emissions using a slurry of:

(A) Sodium chloride
(B) Limestone ($CaCO_3$) or slaked lime [$Ca(OH)_2$], producing synthetic gypsum
(C) Sulfuric acid
(D) Ammonium hydroxide

47. Selective Catalytic Reduction (SCR) converts harmful flue gas nitrogen oxides (NO_x) into harmless N_2 and H_2O using:

(A) Methane injection
(B) Ammonia (NH_3) or urea injection over titanium/vanadium oxide catalysts
(C) Carbon monoxide gas
(D) Sodium bicarbonate

48. The oxygen sag curve in a river downstream of an organic wastewater outfall is modeled by which classical equation?

(A) Streeter-Phelps equation
(B) Michaelis-Menten equation
(C) Navier-Stokes equation
(D) Nernst-Planck equation

49. In the Streeter-Phelps water quality model, the critical dissolved oxygen deficit point represents the location where:

(A) Deoxygenation rate by bacteria exactly equals the atmospheric reaeration rate
(B) All fish die instantly
(C) BOD reaches zero
(D) River velocity becomes supersonic

50. In advanced wastewater treatment, Fenton's Reagent is an Advanced Oxidation Process (AOP) that generates hydroxyl radicals (.OH) using:

(A) Hydrogen peroxide (H_2O_2) catalyzed by ferrous iron (Fe^{2+})
(B) Ozone gas and chlorine
(C) UV light and sodium chloride
(D) Potassium permanganate and lime

51. Advanced Oxidation Processes (AOPs) destroy refractory organic pollutants in wastewater primarily by generating:

(A) Hydroxyl free radicals (.OH) with an extreme standard redox potential of +2.80 V
(B) Sulfate precipitates
(C) Dissolved nitrogen bubbles
(D) Hydrochloric acid mist

52. Activated carbon removes dissolved synthetic organic micropollutants from drinking water primarily by which mechanism?

(A) Chemical covalent reduction
(B) Physical and chemical adsorption onto a microporous internal surface area of 800-1200 m^2/g
(C) Thermal decomposition at room temperature
(D) Magnetic precipitation

53. The correct pore-size sequence of pressure-driven membrane filtration processes from largest to smallest pore rating is:

(A) Microfiltration > Ultrafiltration > Nanofiltration > Reverse Osmosis
(B) Reverse Osmosis > Nanofiltration > Ultrafiltration > Microfiltration
(C) Ultrafiltration > Microfiltration > Reverse Osmosis > Nanofiltration
(D) Nanofiltration > Reverse Osmosis > Microfiltration > Ultrafiltration

54. Which membrane filtration process is capable of separating monovalent dissolved mineral salts (Na^+ and Cl^-) from seawater?

(A) Microfiltration (MF)
(B) Ultrafiltration (UF)
(C) Reverse Osmosis (RO)
(D) Nanofiltration (NF)

55. In oil spill marine response, chemical dispersants (e.g., Corexit) remediate floating surface slicks primarily by:

(A) Neutralizing crude oil into harmless freshwater
(B) Reducing oil-water interfacial tension, breaking slicks into microscopic droplets dispersed into the water column
(C) Polymerizing liquid crude oil into buoyant solid bricks
(D) Igniting the surface oil into a clean flare

56. Which marine hydrocarbonoclastic bacterium plays a preeminent natural role in biodegrading oceanic petroleum alkane spills?

(A) Alcanivorax borkumensis
(B) Escherichia coli
(C) Streptococcus pneumoniae
(D) Bacillus anthracis

57. Adding two independent, equal acoustic sound sources of 80 dB each yields a combined total Sound Pressure Level of:

(A) 160 dB
(B) 83 dB
(C) 86 dB
(D) 90 dB

58. The A-weighting decibel scale [dBA] is used in environmental noise monitoring because it:

(A) Measures pure ultrasonic frequencies above 20 kHz
(B) Filters acoustic frequencies to mimic the human ear's relative frequency sensitivity curve
(C) Amplifies sub-audible infrasound vibrations
(D) Converts sound waves into electrical volts directly

59. In thermoelectric and nuclear power plants, wet evaporative cooling towers dissipate waste condenser heat primarily through:

(A) Direct thermal conduction to dry air alone
(B) Evaporation of a small fraction (1-3%) of circulating water, exploiting water's high latent heat of vaporization
(C) Nuclear transmutation of cooling water
(D) Subterranean geothermal absorption

60. Low-Level Radioactive Waste (LLW), such as contaminated protective laboratory clothing, filters, and mop heads, is conventionally disposed of via:

(A) Deep boreholes 5 km into the Earth's mantle
(B) Engineered shallow land burial trenches
(C) Dumping into open ocean trenches
(D) Vitrification in borosilicate glass

61. The biologically hazardous radioisotope Cesium-137 (half-life = 30.17 years) released in nuclear reactor accidents mimics which physiological element in muscle tissue?

(A) Calcium
(B) Potassium
(C) Iron
(D) Iodine

62. The fission product Strontium-90 (half-life = 28.8 years) poses severe long-term carcinogenic risk because it chemically substitutes for:

(A) Calcium in human bones and teeth, irradiating bone marrow
(B) Sodium in nervous action potentials
(C) Hemoglobin iron in red blood cells
(D) Iodine in the thyroid gland

63. During high-temperature municipal solid waste incineration, incomplete combustion of chlorinated plastics in the presence of copper catalysts generates deadly:

(A) Polychlorinated dibenzo-p-dioxins (PCDDs) and polychlorinated dibenzofurans (PCDFs)
(B) Carbon monoxide and hydrogen gas only
(C) Pure ozone gas
(D) Sulfur hexafluoride

64. To prevent de-novo synthesis of toxic dioxins and furans, modern municipal waste incinerator flue gases must be maintained above:

(A) 400 deg C for 0.1 seconds
(B) 850 to 1100 deg C for at least 2.0 seconds, followed by rapid flue gas quench
(C) 100 deg C for 1 hour
(D) 2000 deg C under high vacuum

65. In municipal aerobic composting, the optimal initial Carbon-to-Nitrogen (C:N) ratio of the mixed waste feed is:

(A) 5:1 to 10:1
(B) 25:1 to 30:1
(C) 80:1 to 100:1
(D) 500:1

66. The high-temperature thermophilic phase (55 to 65 deg C) achieved during managed aerobic windrow composting is critical because it:

(A) Destroys pathogenic bacteria (Salmonella, E. coli) and weed seeds
(B) Converts all carbon into charcoal
(C) Freezes the compost pile into peat
(D) Halts all bacterial decomposition permanently

67. The specialized epigeic earthworm species globally utilized in municipal and agricultural vermicomposting is:

(A) Eisenia fetida (Red Wiggler)
(B) Lumbricus terrestris
(C) Pheretima posthuma
(D) Hirudo medicinalis

68. The Basel Convention on the Control of Transboundary Movements of Hazardous Wastes (1989) mandates that exporting states must obtain:

(A) Prior Informed Consent (PIC) from the designated authority of the importing state
(B) A commercial trade permit from the World Bank
(C) Approval from the International Maritime Organization alone
(D) Zero authorization if shipped by air

69. Printed Circuit Boards (PCBs) from discarded electronic waste (e-waste) represent valuable secondary metal reserves containing high concentrations of:

(A) Copper (approx 20%), Gold, Silver, and Palladium
(B) Uranium and Thorium
(C) Titanium and Magnesium exclusively
(D) Mercury only

70. The environmental policy concept that places financial and physical responsibility for post-consumer e-waste collection and recycling onto manufacturers is:

(A) Extended Producer Responsibility (EPR)
(B) Corporate Tax Exemption
(C) Polluter Pays Retaliation
(D) Precautionary Waiver

71. In wastewater treatment, the primary purpose of adding chemical coagulants such as Alum [Al2(SO4)3] or Ferric Chloride (FeCl3) is to:

(A) Disinfect water from viral pathogens
(B) Neutralize electrostatic negative surface charges on suspended colloidal particles, allowing them to aggregate
(C) Acidify wastewater to pH 2
(D) Dissolve all mineral salts into clear solution

72. The chemical parameter 'Mixed Liquor Volatile Suspended Solids' (MLVSS) in an activated sludge aeration tank measures:

(A) The active biological fraction of the microbial sludge biomass
(B) The total weight of inorganic sand particles
(C) The volume of dissolved oxygen gas
(D) The heavy metal concentration in effluent

73. An Upflow Anaerobic Sludge Blanket (UASB) reactor treats high-strength industrial organic wastewaters by forming dense, highly settleable:

(A) Aerobic fungal mats
(B) Self-aggregated anaerobic microbial granules
(C) Chemical calcium carbonate crusts
(D) Floating oil scums

74. In municipal solid waste management, Refuse-Derived Fuel (RDF) is produced by:

(A) Shredding, drying, and pelletizing combustible waste fractions (paper, textiles, non-recyclable plastics)
(B) Directly burning wet organic food waste
(C) Mixing sewage sludge with radioactive fly ash
(D) Dissolving plastic waste in sulfuric acid

75. The primary toxic chemical component of concern released from improper disposal of cathode ray tube (CRT) computer monitors is:

(A) Lead (Pb) in funnel glass (up to 2-3 kg per CRT monitor)
(B) Beryllium copper springs
(C) Cadmium telluride coatings
(D) Lithium cobalt oxide

76. In a standard laboratory BOD test, 10 mL of raw wastewater is added to a 300 mL BOD incubation bottle (dilution factor = 30). Initial DO = 8.6 mg/L, and DO after 5 days at 20 deg C = 2.6 mg/L. What is the BOD5 of the wastewater?

(A) 60 mg/L
(B) 120 mg/L
(C) 180 mg/L
(D) 240 mg/L

77. For the wastewater in Question 76 with BOD5 = 180 mg/L, calculate the Ultimate Carbonaceous BOD (BOD_u), assuming a deoxygenation rate constant k = 0.23 day^-1 (base e) at 20 deg C, given 1 - exp(-0.23 * 5) = 0.6834:

(A) 180.0 mg/L
(B) 263.4 mg/L
(C) 325.0 mg/L
(D) 400.0 mg/L

78. Industrial biotechnology wastewater contains 460 mg/L of dissolved ethanol (C2H5OH, MW = 46 g/mol). Based on the stoichiometric reaction C2H5OH + 3 O2 -> 2 CO2 + 3 H2O, what is the theoretical Chemical Oxygen Demand (COD)?

(A) 460 mg/L
(B) 960 mg/L
(C) 1,380 mg/L
(D) 1,840 mg/L

79. Fermentation wastewater contains 1,800 mg/L of glucose (C6H12O6, MW = 180 g/mol). Stoichiometric complete oxidation requires 6 moles of O2 (192 g O2) per mole of glucose. What is the theoretical COD of this stream?

(A) 960 mg/L
(B) 1,200 mg/L
(C) 1,920 mg/L
(D) 2,400 mg/L

80. Two identical industrial compressors produce a noise level of 85 dB each. What is the combined total Sound Pressure Level when both operate simultaneously [take log10(2) = 0.301]?

(A) 85.0 dB
(B) 88.0 dB
(C) 170.0 dB
(D) 95.0 dB

81. Three machines generate individual noise levels of 80 dB, 83 dB, and 85 dB. Using logarithmic addition [10^8.0 = 1.0 * 10^8; 10^8.3 = 2.0 * 10^8; 10^8.5 = 3.16 * 10^8; log10(6.16) = 0.7896], calculate total Sound Pressure Level:

(A) 85.2 dB
(B) 87.9 dB
(C) 92.5 dB
(D) 98.0 dB

82. An Electrostatic Precipitator (ESP) treats a flue gas flow rate Q = 100 m3/s. The total collection plate area A = 5,000 m2 and fly ash particle migration velocity w = 0.08 m/s. Using the Deutsch-Anderson equation [eta = 1 - exp(-w*A/Q)], calculate collection efficiency (given exp(-4.0) = 0.0183):

(A) 95.00%
(B) 98.17%
(C) 99.50%
(D) 99.99%

83. A point acoustic noise source generates 90 dB at a distance of 10 meters in a hemispherical free field. According to the inverse square law [Delta L = 20 * log10(r2 / r1)], what is the sound level at a distance of 40 meters [given log10(4) = 0.602]?

(A) 72 dB
(B) 78 dB
(C) 84 dB
(D) 88 dB

84. Radioactive Iodine-131 has a radiological half-life of 8.0 days. What fraction of original activity remains in biological tissue after 32 days (4 half-lives)?

(A) 12.5%
(B) 6.25%
(C) 3.125%
(D) 1.56%

85. Cesium-137 has a half-life of 30.17 years (9.52 * 10^8 seconds). Its radioactive decay constant lambda = ln(2) / t_1/2 is 7.28 * 10^-10 s^-1. What is the initial activity (Bq) of 1.0 milligram of pure Cs-137 (4.39 * 10^18 atoms)?

(A) 3.20 * 10^6 Bq
(B) 3.20 * 10^9 Bq (3.20 GBq)
(C) 3.20 * 10^12 Bq
(D) 1.00 * 10^3 Bq

86. An activated sludge aeration basin of volume V = 2,500 m3 treats an influent flow rate Q = 10,000 m3/day with influent BOD5 S0 = 200 mg/L (200 g/m3). The mixed liquor suspended solids MLSS X = 2,000 mg/L (2,000 g/m3). Calculate the Food-to-Microorganism (F/M) ratio in day^-1:

(A) 0.20 day^-1
(B) 0.40 day^-1
(C) 0.80 day^-1
(D) 1.20 day^-1

87. For the activated sludge plant in Question 86, the total microbial biomass in the aeration basin is 5,000 kg. If the waste activated sludge mass discharged is 500 kg/day, what is the Mean Cell Residence Time (Sludge Age, theta_c)?

(A) 5 days
(B) 10 days
(C) 15 days
(D) 20 days

88. A municipality generates 500 metric tons of municipal solid waste per day. If compacted landfill waste density is 800 kg/m3 (0.8 tons/m3) and 20% soil daily cover is added by volume, what is the total landfill volume required annually (365 days)?

(A) 136,875 m3
(B) 228,125 m3
(C) 273,750 m3
(D) 350,000 m3

89. A regional sanitary landfill has a total engineered capacity of 2,737,500 m3. Based on the annual filling rate in Question 88 (273,750 m3/year), what is the projected design lifespan of the landfill?

(A) 5 years
(B) 10 years
(C) 15 years
(D) 25 years

90. A 10-hectare (100,000 m2) landfill receives 1,200 mm (1.2 m) annual precipitation. Surface runoff coefficient is 0.25 (25%), and actual evapotranspiration is 0.50 (50%). What is the estimated annual volume of leachate generated by net precipitation infiltration?

(A) 15,000 m3/yr
(B) 30,000 m3/yr
(C) 60,000 m3/yr
(D) 90,000 m3/yr

91. In a standard cyclone separator, the 50% cut diameter d50 is inversely proportional to the square root of the inlet gas velocity (v_i). If gas inlet velocity v_i is quadrupled (4x), the particle cut diameter d50:

(A) Doubles
(B) Halves (reduces by factor of 2)
(C) Quadruples
(D) Reduces by factor of 16

92. Flue gas from a coal boiler contains 2,000 mg/m3 of SO2. A wet limestone scrubber operates at 95% removal efficiency. What is the concentration of SO2 discharged from the chimney stack?

(A) 50 mg/m3
(B) 100 mg/m3
(C) 200 mg/m3
(D) 500 mg/m3

93. A thermal power plant emits 10 metric tons of SO2 per day (156.25 kmol SO2/day). In a wet scrubber: SO2 + CaCO3 + 0.5 O2 + 2 H2O -> CaSO4.2H2O. What mass of pure limestone (CaCO3, MW = 100 g/mol) is stoichiometrically required per day?

(A) 10.00 metric tons
(B) 15.63 metric tons
(C) 20.50 metric tons
(D) 31.25 metric tons

94. A batch of 10 metric tons of discarded computer printed circuit boards contains 250 grams of gold (Au) per ton. If a hydrometallurgical cyanide-free thiosulfate leaching process achieves 90% recovery, what mass of pure gold is recovered?

(A) 1,250 grams
(B) 2,000 grams
(C) 2,250 grams (2.25 kg)
(D) 2,500 grams

95. A secondary municipal wastewater effluent flow of 50,000 m3/day is disinfected with chlorine. The required chlorine dosage is 8.0 mg/L (8.0 g/m3). What mass of pure chlorine gas (kg/day) must be fed into the contact chamber daily?

(A) 200 kg/day
(B) 400 kg/day
(C) 800 kg/day
(D) 1,200 kg/day

96. In a jar test for water treatment, optimal coagulation of raw river water occurs at an alum dosage of 25 mg/L. To treat a water supply of 40 million liters per day (40 MLD), what mass of alum is required daily?

(A) 500 kg/day
(B) 1,000 kg/day (1.0 metric ton)
(C) 2,500 kg/day
(D) 4,000 kg/day

97. An industrial worker is exposed to an 8-hour noise regime: 4 hours at 90 dBA (permissible OSHA time = 8 hours) and 2 hours at 95 dBA (permissible time = 4 hours). Calculate the occupational noise dose D = 100 * [C1/T1 + C2/T2]%:

(A) 50%
(B) 75%
(C) 100% (maximum permissible 8-hr dose)
(D) 150%

98. A seawater Reverse Osmosis (SWRO) desalination unit is fed with 100 m3/hr of raw seawater and produces 45 m3/hr of freshwater permeate. What is the recovery rate of the membrane system?

(A) 35.0%
(B) 45.0%
(C) 55.0%
(D) 75.0%

99. For the SWRO system in Question 98, feed seawater TDS is 35,000 mg/L and permeate product TDS is 350 mg/L. What is the membrane salt rejection percentage [R = (1 - C_p/C_f) * 100]?

(A) 90.0%
(B) 95.0%
(C) 99.0%
(D) 99.9%

100. Lead has a Half-Value Layer (HVL) thickness of 1.2 cm for 1.0 MeV gamma rays. What thickness of lead shield is required to attenuate radiation transmission to 1/16 (6.25%) of its unshielded incident intensity?

(A) 2.4 cm
(B) 3.6 cm
(C) 4.8 cm
(D) 6.0 cm

SECTION B: DESCRIPTIVE & SUBJECTIVE QUESTIONS

Q1. Detail the atmospheric chemical mechanisms governing Photochemical Smog formation, formulating the cyclic reactions of NOx, VOCs, ozone, and PAN.

Q2. Contrast Classical (London-type) Smog with Photochemical (Los Angeles-type) Smog in terms of chemistry, meteorology, and pathology.

Q3. Explain the atmospheric oxidation chemistry of Sulfur Dioxide and Nitrogen Oxides generating Acid Rain, and detail its impact on marble structures.

Q4. Formulate the physical operating principles of an Electrostatic Precipitator (ESP), derive the Deutsch-Anderson collection efficiency, and analyze resistivity.

Q5. Explain the catalytic chemistry of Automotive Three-Way Catalytic Converters, detailing the role of Pt, Pd, Rh, and the closed-loop lambda oxygen sensor.

Q6. Detail the chemistry and engineering design of Flue Gas Desulfurization (FGD) wet limestone scrubbers, including gypsum forced oxidation.

Q7. Compare the analytical methodologies and biochemical kinetics of Biochemical Oxygen Demand (BOD5) and Chemical Oxygen Demand (COD).

Q8. Derive the Streeter-Phelps Dissolved Oxygen (DO) Sag Equation, and determine the critical oxygen deficit time (tc) and location (xc).

Q9. Explain the microbiological ecology and engineering design of the Activated Sludge Process (ASP), defining F/M ratio and Sludge Age.

Q10. Detail the operational mechanics of an Upflow Anaerobic Sludge Blanket (UASB) reactor, including Granulation and the Gas-Liquid-Solid (GLS) separator.

Q11. Explain the chemical reaction mechanisms of Advanced Oxidation Processes (AOPs), focusing on Fenton's Reagent, Photo-Fenton, and Ozone/UV systems.

Q12. Analyze the molecular toxicology and cellular pathology of Mercury (Minamata) and Cadmium (Itai-Itai) poisoning in environmental systems.

Q13. Examine the geochemical origins of Arsenic and Fluoride groundwater contamination in India, detailing the Nalgonda water defluoridation technique.

Q14. Analyze the physicochemical weathering of Marine Oil Spills, and evaluate mechanical containment and bioremediation by Alcanivorax borkumensis.

Q15. Formulate the acoustic physics of Noise Pollution, decibel addition, equal loudness contours, and occupational sensorineural hearing conservation.

Q16. Analyze the physical thermodynamics of Thermal Pollution from thermoelectric facilities, Henry's Law gas solubility decline, and cooling tower engineering.

Q17. Detail the biophysics of Ionizing Radiation (Alpha, Beta, Gamma), formulate Linear Energy Transfer (LET), and explain double-strand DNA damage.

Q18. Analyze the reactor physics and environmental aftermath of the Chernobyl Nuclear Disaster (1986), detailing the Positive Void Coefficient.

Q19. Examine the sequence of events during the Fukushima Daiichi Nuclear Disaster (2011), detailing Zircaloy-Water reactions and hydrogen explosions.

Q20. Detail the Multi-Barrier Concept for High-Level Radioactive Waste (HLW) isolation, explaining Borosilicate Glass Vitrification and deep repositories.

Q21. Formulate the Integrated Solid Waste Management (ISWM) hierarchy, evaluating the 5Rs and municipal waste flowsheets.

Q22. Detail the engineering design of a Modern Sanitary Landfill, including multi-layer composite liners, leachate collection, and methane gas recovery.

Q23. Explain the thermochemical combustion stages of Waste-to-Energy Incineration, flue gas scrubbing, and prevention of Dioxins and Furans (PCDD/Fs).

Q24. Compare Aerobic Windrow Composting with Vermicomposting, formulating C:N ratio kinetics, microbial succession, and Eisenia fetida physiology.

Q25. Analyze thermochemical Pyrolysis and Gasification of Solid Wastes, comparing syngas production with direct incineration and plasma gasification.

Q26. Examine the regulatory classification of Hazardous Waste under the US EPA and Indian Rules: Ignitability, Corrosivity, Reactivity, and Toxicity.

Q27. Detail the Indian Bio-Medical Waste Management Rules (2016), specifying the fourfold color-coded segregation, autoclaving, and treatment pathways.

Q28. Explain hydrometallurgical and bioleaching metallurgical extraction processes for recovering Gold, Silver, and Copper from Electronic Waste.

Q29. Detail the Extended Producer Responsibility (EPR) policy architecture for E-Waste and Plastic Waste, including digital EPR certificate trading.

Q30. Formulate the engineering architecture of Industrial Wastewater Zero Liquid Discharge (ZLD), detailing thermal vapor recompression and crystallization.

SECTION C: ANSWER KEY - MULTIPLE CHOICE QUESTIONS

Q1:	[A]	**Q21:**	[B]	**Q41:**	[A]	**Q61:**	[B]	**Q81:**	[B]
Q2:	[C]	**Q22:**	[B]	**Q42:**	[B]	**Q62:**	[A]	**Q82:**	[B]
Q3:	[B]	**Q23:**	[B]	**Q43:**	[B]	**Q63:**	[A]	**Q83:**	[B]
Q4:	[B]	**Q24:**	[B]	**Q44:**	[B]	**Q64:**	[B]	**Q84:**	[B]
Q5:	[A]	**Q25:**	[B]	**Q45:**	[A]	**Q65:**	[B]	**Q85:**	[B]
Q6:	[B]	**Q26:**	[B]	**Q46:**	[B]	**Q66:**	[A]	**Q86:**	[B]
Q7:	[B]	**Q27:**	[A]	**Q47:**	[B]	**Q67:**	[A]	**Q87:**	[B]
Q8:	[A]	**Q28:**	[B]	**Q48:**	[A]	**Q68:**	[A]	**Q88:**	[C]
Q9:	[B]	**Q29:**	[A]	**Q49:**	[A]	**Q69:**	[A]	**Q89:**	[B]
Q10:	[B]	**Q30:**	[A]	**Q50:**	[A]	**Q70:**	[A]	**Q90:**	[B]
Q11:	[B]	**Q31:**	[A]	**Q51:**	[A]	**Q71:**	[B]	**Q91:**	[B]
Q12:	[B]	**Q32:**	[B]	**Q52:**	[B]	**Q72:**	[A]	**Q92:**	[B]
Q13:	[A]	**Q33:**	[C]	**Q53:**	[A]	**Q73:**	[B]	**Q93:**	[B]
Q14:	[B]	**Q34:**	[B]	**Q54:**	[C]	**Q74:**	[A]	**Q94:**	[C]
Q15:	[B]	**Q35:**	[B]	**Q55:**	[B]	**Q75:**	[A]	**Q95:**	[B]
Q16:	[B]	**Q36:**	[B]	**Q56:**	[A]	**Q76:**	[C]	**Q96:**	[B]
Q17:	[B]	**Q37:**	[B]	**Q57:**	[B]	**Q77:**	[B]	**Q97:**	[C]
Q18:	[B]	**Q38:**	[B]	**Q58:**	[B]	**Q78:**	[B]	**Q98:**	[B]
Q19:	[B]	**Q39:**	[B]	**Q59:**	[B]	**Q79:**	[C]	**Q99:**	[C]
Q20:	[B]	**Q40:**	[C]	**Q60:**	[B]	**Q80:**	[B]	**Q100:**	[C]

MCQ Scoring Guide & Difficulty Distribution:

Questions 1 to 40: Foundational & Conceptual [Easy] | Questions 41 to 75: Analytical & Applied [Moderate]

Questions 76 to 100: Complex Numerical & Advanced Problem Solving [Difficult]

* Los Angeles Smog: Chemically OXIDIZING smog dominated by secondary oxidants (ground-level ozone O3, NO2, Peroxyacetyl nitrate PAN, aldehydes).
- Parameter: Primary Anthropogenic Source
* London Smog: Domestic open-hearth burning of bituminous coal and industrial coal-fired power stations.
* Los Angeles Smog: Exhaust emissions from internal combustion engine vehicles (gasoline/diesel traffic) and industrial solvents.
- Parameter: Meteorological Pre-conditions
* London Smog: Cold winter temperatures (0 to 5 deg C), stagnant radiation temperature inversions, high relative humidity, and dense fog.
* Los Angeles Smog: Warm summer/autumn temperatures (> 25 to 35 deg C), intense ultraviolet (UV) solar radiation, low humidity, and subsidence inversions.
- Parameter: Peak Time of Diurnal Occurrence
* London Smog: Early morning before sunrise (winter fog trapped near ground).
* Los Angeles Smog: Midday to early afternoon (12:00 to 15:00 hours) when cumulative solar UV flux reaches its daily peak.
- Parameter: Human Pathology
* London Smog: Acute bronchial spasms, acute pulmonary edema, fatal chronic bronchitis in elderly individuals.
* Los Angeles Smog: Intense eye irritation (lachrymatory action of PAN), respiratory tightness, impaired athletic lung capacity, foliar necrosis in crops.

>>**Solution to Question 3:** Explain the atmospheric oxidation chemistry of Sulfur Dioxide and Nitrogen Oxides generating Acid Rain, and detail its impact on marbl

$CaCO_3 (s) + H_2SO_4 (aq) + H_2O \rightarrow CaSO_4.2H_2O (s)$ (Gypsum) + $CO_2 (g)$

- Chemical Pathology: Calcium carbonate is converted into calcium sulfate dihydrate (gypsum).
- Volume Expansion & Spalling: Gypsum has a molar volume nearly double that of calcite. As crystals form in micro-fissures, internal crystallographic stress causes the marble surface to blister, flake, and undergo crumbling (termed 'Marble Cancer'), compounded by soot discoloration.

>> Solution to Question 5: Formulate the physical operating principles of an Electrostatic Precipitator (ESP), derive the Deutsch-Anderson collection efficiency, an

[Problem Statement]: Explain the catalytic chemistry of Automotive Three-Way Catalytic Converters, detailing the role of Pt, Pd, Rh, and the close
[Detailed Step-by-Step Answer]:
Automotive Three-Way Catalytic Converters: Catalyst Metallurgy, Stoichiometry, and Closed-Loop Control:
1. The Three Primary Regulated Pollutants:
Internal combustion gasoline engines emit three hazardous exhaust gases:
- Carbon Monoxide (CO): Toxic product of incomplete fuel combustion.
- Unburned Hydrocarbons (HC / VOCs): Fuel molecules escaping quench zones.
- Nitrogen Oxides (NO and NO2, collectively NOx): Formed by high-temperature thermal fixation of atmospheric N2 and O2 in the combustion chamber.
2. Catalyst Structure and Active Noble Metals:
- Monolithic Substrate: A ceramic cordierite (2MgO.2Al2O3.5SiO2) honeycomb structure extruded with 400 to 600 micro-channels per square inch to provide massive geometric contact area with minimal exhaust backpressure.
- Washcoat: A porous gamma-Al2O3 layer containing ceria-zirconia (CeO2-ZrO2) as an Oxygen Storage Component (OSC).
- Active Catalyst Loading:
* Platinum (Pt) and Palladium (Pd): Oxidation catalysts.
* Rhodium (Rh): Reduction catalyst.
3. Simultaneous Heterogeneous Catalytic Reactions:
- Reduction Stage (Catalyzed by Rhodium):
2 NO + 2 CO --[Rh]--> N2 + 2 CO2
2 NO + 2 H2 --[Rh]--> N2 + 2 H2O
NOx is selectively reduced to harmless diatomic nitrogen (N2).
- Oxidation Stage (Catalyzed by Platinum and Palladium):
2 CO + O2 --[Pt/Pd]--> 2 CO2
C_x H_y + (x + y/4) O2 --[Pt/Pd]--> x CO2 + (y/2) H2O
Carbon monoxide and unburned hydrocarbons are fully oxidized to carbon dioxide and water vapor.
4. Closed-Loop Lambda (lambda) Oxygen Sensor Feedback:
- High simultaneous conversion (> 95%) of ALL THREE pollutants occurs exclusively within an extremely narrow air-to-fuel ratio window: lambda = 1.000 +/- 0.005 (Stoichiometric AFR = 14.7 kg air per kg gasoline).
- If mixture is Lean (lambda > 1, excess oxygen): CO and HC are oxidized, but NOx reduction efficiency plummets.
- If mixture is Rich (lambda < 1, oxygen deficit): NOx is reduced, but CO and HC oxidation collapses.
- The zirconia oxygen sensor (lambda sensor) in the exhaust pipe continuously measures residual O2 partial pressure, feeding real-time millivolt signals to the Engine Control Unit (ECU) to adjust electronic fuel injectors hundreds of times per minute.

>>[Problem Statement]: **Solution Statement Question 6:** Detail the chemistry and engineering design of Flue Gas Desulfurization (FGD) wet limestone scrubbers, including gypsum forced oxida

SO2 (g) <-> SO2 (aq)
SO2 (aq) + H2O <-> H2SO3 <-> H+ + HSO3- (Bisulfite)
- Step 2: Limestone Dissolution:
Finely ground limestone (calcium carbonate, CaCO3, pulverized to 90% < 44 microns) dissolves in the acidic slurry:
CaCO3 (s) + H+ <-> Ca2+ + HCO3-
- Step 3: Calcium Sulfite Precipitation:
Ca2+ + HSO3- + 1/2 H2O -> CaSO3.1/2H2O (s) v + H+
Forms calcium sulfite hemihydrate flocs.
- Step 4: Forced In-Situ Oxidation (Compressed Air Injection):
In the bottom reaction tank of the scrubber, compressed air is vigorously sparged into the slurry to fully oxidize sulfite into sulfate:
HSO3- + 1/2 O2 -> SO4^2- + H+
Ca2+ + SO4^2- + 2 H2O -> CaSO4.2H2O (s) v (Gypsum).
3. Overall Net Stoichiometric Reaction:
SO2 (g) + CaCO3 (s) + 1/2 O2 (g) + 2 H2O (l) -> CaSO4.2H2O (s) + CO2 (g)
4. By-Product Commercial Valorization:
The precipitated calcium sulfate dihydrate (synthetic gypsum) is dewatered via vacuum hydrocyclones and belt filters to < 10% moisture, yielding high-purity (> 95%) commercial wallboard gypsum used in construction, eliminating solid waste landfill disposal.

>>**Solution to Question 7:** Compare the analytical methodologies and biochemical kinetics of Biochemical Oxygen Demand (BOD5) and Chemical Oxygen Dema

- Industrial Effluents (Petrochemical, Textile, Pharmaceutical): COD/BOD5 ratio > 3.0 to 5.0 indicates the presence of toxic or refractory organic compounds (dyes, phenols, surfactants) requiring chemical or advanced oxidation pre-treatment before biological treatment.

>>Solution Step Question Solve the Streeter-Phelps Dissolved Oxygen (DO) Sag Equation. Activated Sludge Process (ASP) definition, Flow diagram, and Sludge Age.

maintain high active biocatalyst concentration.
- Waste Activated Sludge (WAS) Line: Discharges excess synthesized microbial biomass to control sludge age.
2. Microbial Floc Ecology:
The functional unit is the Activated Sludge Floc: an agglomeration of heterotrophic bacteria (Zoogloea ramigera, Pseudomonas), encapsulated within an extracellular polymeric substance (EPS) matrix composed of polysaccharides, proteins, and DNA. Ciliated protozoa (Vorticella, Epistylis) graze on free-swimming dispersed bacteria, clarifying the effluent water.
3. Crucial Engineering Operational Parameters:
- Food-to-Microorganism (F/M) Ratio:
Quantifies the organic loading rate per unit of microbial biomass per day:
F/M = (Q * S_0) / (V * X)
Where Q is influent flow (m3/day), S_0 is influent BOD5 (mg/L), V is aeration tank volume (m3), and X is MLSS (mg/L).
* Standard Conventional ASP operates at F/M = 0.2 to 0.4 day^-1.
* High F/M (> 0.6) leads to incomplete BOD removal and turbid effluent.
* Low F/M (< 0.1, Extended Aeration) promotes endogenous respiration and highly stable, well-settling sludge.
- Mean Cell Residence Time (Sludge Age / Solids Retention Time, theta_c):
The average time that microbial biomass remains in the system:
theta_c = (V * X) / (Q_w * X_w + Q_e * X_e)
Where Q_w is waste sludge flow rate, X_w is waste sludge concentration, and Q_e * X_e is effluent suspended solids loss.
* Conventional ASP maintains theta_c = 5 to 15 days.
* Nitrification requires theta_c > 10 to 12 days to prevent washout of slow-growing autotrophic nitrifiers (Nitrosomonas and Nitrobacter).

>>**Solution to Question 18:** the operational mechanics of an Upflow Anaerobic Sludge Blanket (UASB) reactor, including Granulation and the Gas-Liquid-Sol

inverted settlement baffles and gas collection hoods:
- Gas Separation: Upward-rising biogas bubbles (CH4, CO2) strike the inclined bottom hoods and are deflected into the gas collection dome, preventing biogas from disrupting the upper clarification zone.
- Solid Settlement: Degassed sludge flocs enter the calm quiescent settler zone above the baffles. As liquid velocity drops, the dense granules settle under gravity back down into the lower digestion blanket.
- Clarified Liquid Discharge: The clean liquid effluent overflows into top peripheral collection launders.

- Chemical Speciation & Lipophilicity: While elemental mercury (Hg0) and inorganic mercuric salts (Hg2+) have limited absorption across biological membranes, biological methylation by anoxic sediment bacteria produces monomethylmercury $[CH_3Hg]^+$.
- Molecular Mimicry and Transport:
Methylmercury possesses an extraordinary chemical affinity for sulfhydryl (-SH) thiol groups on L-cysteine. It forms an organometallic coordination complex: $[CH_3Hg\text{-}S\text{-}CH_2\text{-}CH(NH_2)\text{-}COOH]$ (Methylmercury-cysteine).
Because this complex is structurally and stereochemically analogous to the essential amino acid L-methionine, the endothelial L-type amino acid transporter 1 (LAT1) actively transports it across the blood-brain barrier and placenta into the central nervous system.
- Cellular Pathology: Inside neurons, methylmercury binds to glutathione, depletes antioxidant defenses, generates massive Reactive Oxygen Species (ROS), and inhibits tubulin polymerization (disrupting mitotic spindles and axonal transport). Pathological hallmark: Bilateral cerebral cortical atrophy (focal necrosis of calcarine cortex and granule cell layer of cerebellum), manifesting as severe sensory ataxia, concentric constriction of visual fields (tunnel vision), hearing loss, dysarthria, paralysis, and severe microcephaly in developing fetuses.
2. Cadmium Poisoning (Itai-Itai Disease):
- Source & Environmental Pathway: Industrial mining and smelting effluent discharged into the Jinzu River basin in Toyama, Japan, contaminated downstream paddy fields. Rice plants bio-accumulated cadmium to high levels.
- Renal Proximal Tubular Pathology:
Absorbed Cd^{2+} induces hepatic synthesis of Metallothionein (MT). The Cd-MT complex is filtered by renal glomeruli and reabsorbed by proximal convoluted tubule cells. In lysosomes, MT is degraded, releasing free Cd^{2+} that poisons renal ATPases and enzymes.
* Fanconi Syndrome: Irreversible proximal tubular damage impairs tubular reabsorption, causing massive proteinuria, glucosuria, aminoaciduria, and hypercalciuria (urinary calcium wasting).
- Osteomalacia and Bone Collapse:
Cadmium inhibits renal 25-hydroxyvitamin D 1-alpha-hydroxylase, preventing synthesis of active 1,25-dihydroxyvitamin D3 (calcitriol). Hypocalcemia and impaired intestinal calcium absorption force the parathyroid glands to mobilize calcium from skeletal bones. Bones become severely osteomalacic, fragile, and undergo multiple spontaneous fractures upon minimal mechanical stress ('Itai-Itai' = 'It hurts! It hurts!').

>>Solution to Question 13: Examine the geochemical origins of Arsenic and Fluoride groundwater contamination in India, detailing the Nalgonda water defluoridati

2. Geochemical Origins of Fluoride Contamination:
- Endemic across 19 Indian states (Rajasthan, Gujarat, Andhra Pradesh, Telangana).
- Geogenic Dissolution: Weathering of fluorine-bearing igneous and metamorphic minerals (Fluorite CaF2, Fluorapatite Ca5(PO4)3F, Cryolite Na3AlF6) in granitic aquifers under alkaline, low-calcium groundwater conditions:
CaF2 (s) + 2 NaHCO3 (aq) <-> CaCO3 (s) + 2 Na+ + 2 F- + H2O + CO2.
3. The Nalgonda Defluoridation Technique (NEERI, Nagpur):
- Engineering Protocol: Developed in 1974 by the National Environmental Engineering Research Institute (NEERI) for rural decentralized water defluoridation:
- Sequence of Chemical Additions:
* Step 1: Raw water containing excess fluoride is dosed with Lime [CaO or Ca(OH)2] to provide basic alkalinity and facilitate flocculation.
* Step 2: Dosing with Alum [Aluminum sulfate, Al2(SO4)3.18H2O] under rapid mechanical flash mixing (1-2 minutes).
* Step 3: Slow flocculation mixing (10-15 minutes), followed by gravity sedimentation (1 hour) and filtration through a sand bed.
- Chemical Mechanism:
Alum hydrolyzes into complex polyhydroxy-aluminum species and amorphous Al(OH)3 flocs:
Al3+ + 3 H2O <-> Al(OH)3 (am) + 3 H+
Fluoride ions (F-) possess an ionic radius (1.33 A) identical to the hydroxyl ion (OH-, 1.33 A). In a rapid ligand exchange reaction, fluoride replaces hydroxyl groups in the aluminum hydroxide lattice or is co-precipitated:
Al(OH)3 (s) + x F- -> Al(OH)_{3-x} F_x (s) + x OH-
The dense, fluoride-laden aluminum flocs settle out, reducing fluoride from 10 mg/L down to safe drinking levels (< 1.0 mg/L).

>>**Solution to Question 14:** Analyze the physicochemical weathering of Marine Oil Spills, and evaluate mechanical containment and bioremediation by Alcanivorax

- Alcanivorax borkumensis: An obligate marine hydrocarbon-degrading bacterium that typically comprises < 1% of baseline microbial communities, but blooms to comprise over 80-90% of the microbial population within days of a petroleum spill!
- Catabolic Physiology:
* A. borkumensis exclusively utilizes aliphatic hydrocarbons (C5 to C32 n-alkanes) as its carbon and energy source.
* Genome encodes two alkane hydroxylases (AlkB1 and AlkB2) and three flavin-binding monooxygenases (AlmA).
* Synthesizes and secretes anionic glycolipid biosurfactants that emulsify thick oil slicks into microscopic droplets, exponentially increasing the oil-water interfacial surface area for enzymatic catabolism.
- Biostimulation Interventions:
Open ocean biodegradation is constrained by nitrogen and phosphorus deficits. Applying oleophilic liquid fertilizers (e.g., Inipol EAP-22, urea encapsulated in oleic acid) provides N and P directly at the hydrophobic oil-water interface, accelerating biological cleanup.

>>Solution to Question 15: Formulate the acoustic physics of Noise Pollution, decibel addition, equal loudness contours, and occupational sensorineural hearing c

(lambda < 0.01 nm) emitted by excited nuclear isomers (Co-60, Cs-137). High penetration (requires thick dense lead or reinforced concrete shielding).

2. Linear Energy Transfer (LET) and Relative Biological Effectiveness (RBE):

- Linear Energy Transfer (LET): The average energy deposited per unit distance along the track of the ionizing particle:

LET = -dE / dx [expressed in keV/micrometer].

* Low-LET Radiation (Gamma rays, X-rays, beta particles): LET ~ 0.2 to 2 keV/um. Generates widely spaced, sparse ionization events.
* High-LET Radiation (Alpha particles, heavy recoiling nuclei): LET ~ 100 to 200 keV/um. Generates dense, concentrated ionization tracks.

- Relative Biological Effectiveness (RBE): The ratio of biological damage produced by a test radiation compared to reference 250 kVp X-rays. For alpha radiation, radiation weighting factor w_R = 20 (meaning 1 Gray of alpha dose delivers 20 Sieverts of biological damage).

3. Molecular Radio-Pathology:

- Water Radiolysis: Ionization of intracellular water generates high fluxes of reactive hydroxyl radicals:

H2O + Radiation -> H2O+ + e- -> .OH + H+ + e_aq-.

- Clustered DNA Lesions: While low-LET radiation induces simple repairable single-strand breaks (SSBs), high-LET alpha particles deposit localized energy clusters directly across the DNA double helix. This causes complex, clustered Double-Strand DNA Breaks (DSBs) where both complementary strands are cleaved within a few base pairs. Non-Homologous End Joining (NHEJ) repair fails, leading to chromosomal translocations, mitotic catastrophe, apoptosis, and leukemia.

>> **Solution to Question 18:** Analyze the reactor physics and environmental aftermath of the Chernobyl Nuclear Disaster (1986), detailing the Positive Void Coefficie

the 2,000-ton upper reactor shield (Lid Elena).

- The graphite moderator ignited into a fierce, unquenchable fire burning for 10 days, releasing over 5.2 million Terabecquerels (TBq) of volatile radionuclides into the upper troposphere:

* Iodine-131 (1,760 PBq, half-life = 8 days): Inhaled and ingested in milk, accumulating in children's thyroid glands, causing over 6,000 documented pediatric thyroid carcinomas.
* Cesium-137 (85 PBq, half-life = 30.17 years): Contaminated over 200,000 km2 across Belarus, Ukraine, and Russia, necessitating the permanent 30-km Chernobyl Exclusion Zone.

[Detailed Step-by-Step Answer]:
Multi-Barrier Engineering Architecture for High-Level Radioactive Waste (HLW):
1. Radiological Nature of High-Level Waste:
Spent nuclear reactor fuel consists of unburned uranium (95%), transuranic actinides (Pu-239, Np-237, Am-241 with half-lives exceeding tens of thousands of years), and short-to-medium-lived fission products (Cs-137, Sr-90, Tc-99, I-129). These wastes emit intense thermal heat and lethal gamma radiation, requiring permanent geological isolation for > 100,000 years.
2. Chemical Vitrification into Borosilicate Glass:
- Liquid HLW nitric acid extract is concentrated by evaporation, calcined at 500-800 deg C into solid dry oxides, and mixed with borosilicate glass frit (SiO_2: 50-60%, B_2O_3: 15-20%, Na_2O: 10-15%).
- Melting: The mixture is melted inside an induction-heated ceramic melter at 1,150 deg C. The radioactive polyvalent metal ions (actinides, lanthanides, transition metals) become chemically incorporated into the random network covalent silicon-oxygen-boron structural backbone.
- Properties of Vitrified Waste:
* Exceptional chemical durability: Leaching rate in groundwater is less than 10^{-7} g/cm2/day.
* High thermal conductivity and thermal stability up to 500 deg C.
* High radiation tolerance: Amorphous glass matrix accommodates radiation displacement damage without swelling or macroscopic cracking.
3. Multi-Barrier Geological Isolation (KBS-3 Paradigm):
Vitrified canisters are emplaced in a Deep Geological Repository (DGR) excavated 500 meters underground within stable crystalline rock:
- Engineered Barrier 1 (Vitrified Waste Matrix): Chemically locks rad-ions in glass.
- Engineered Barrier 2 (Overpack Canister): High-purity corrosion-resistant thick Copper (50 mm) or Titanium shell with a cast-iron inner structural support sleeve designed to withstand 50 MPa hydrostatic and lithostatic pressure without mechanical failure for 100,000 years.
- Engineered Barrier 3 (Compacted Sodium Bentonite Clay): Annular space around canisters is packed with dense swelling montmorillonite clay. When hydrated by groundwater, bentonite swells tightly against the rock, providing extremely low hydraulic permeability (< 10^{-12} m/s), plastic self-healing of fractures, and chemical sorption of migrating cations.
- Natural Geological Barrier 4 (Deep Host Bedrock): 500 meters of tectonically stable crystalline granite or bedded salt formations with zero regional groundwater movement.

>>**Solution to Question 21:** Formulate the Integrated Solid Waste Management (ISWM) hierarchy, evaluating the 5Rs and municipal waste flowsheets.

feedstocks.
- 5. Recover (Energy Recovery): Utilizing non-recyclable combustible waste fractions for energy recovery through biological anaerobic digestion (biogas) or thermal waste-to-energy incineration (electricity and steam).
- Least Preferred Baseline: Disposal: Uncontrolled open dumping is prohibited; sanitary landfilling is strictly reserved for inert, non-reactive residuals and combustion ash.
3. Modern City Integrated Waste Flowsheet:
- Source Segregation: Mandatory domestic segregation into Three Streams: Wet Organic Biodegradables (Green Bin), Dry Recyclables (Blue Bin), and Hazardous/Domestic Sanitary Waste (Red/Black Bin).
- Decentralized Processing: Wet organics divert directly to local Aerobic Windrow Composting or Biomethanation plants.
- Material Recovery Facility (MRF): Dry recyclables pass through automated optical sorters, eddy-current separators (non-ferrous aluminum), and magnetic drums (ferrous steel) for high-purity recycling.
- Residual Fraction: Residual high-calorific rejects form Refuse-Derived Fuel (RDF) co-processed in cement kilns, while non-reactive mineral residues enter engineered sanitary landfills.

>>Solution to Question 22: Detailed the engineering design of a Modern Sanitary Landfill, including multi-layer composite liners, leachate collection, and methane gas

header pipe system extract LFG under continuous suction.
- Methane is passed through moisture knock-out drums, desulfurized over iron sponge filters, and combusted in internal combustion gas engines to generate base-load electrical power.

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* 1. Mesophilic Phase (25 to 40 deg C, Days 1-3): Fast-growing mesophilic bacteria (Bacillus, Pseudomonas) and fungi rapidly metabolize soluble, easily digestible sugars and amino acids, releasing heat.
* 2. Thermophilic Phase (50 to 65 deg C, Weeks 1-4): As temperature rises, thermophilic spore-forming bacteria (Thermus, Geobacillus) and actinomycetes take over, degrading complex polymers (starches, cellulose, hemicellulose).
Temperatures >= 55 deg C for at least 15 consecutive days are mandated to kill enteric viral/bacterial pathogens (Salmonella, Shigella) and weed seeds.
* 3. Cooling and Maturation Phase (40 to 25 deg C, Weeks 5-12): Temperature declines. Fungi and actinomycetes synthesize polycondensed aromatic humic and fulvic acids, producing mature, earthy humus with C:N ratio < 15:1.
2. Vermicomposting with Eisenia fetida (Red Wiggler Earthworms):
- Epigeic Earthworm Physiology:
Eisenia fetida is an epigeic litter-dwelling earthworm that does not burrow deep into mineral soils; it feeds voraciously on decaying organic matter at surface temperatures between 15 and 25 deg C.
- Gizzard Mastication and Gut Microbiome:
* Earthworms lack teeth; food is mechanically ground with ingested mineral quartz particles inside the muscular gizzard.
* The worm gut functions as a natural bioreactor rich in symbiotic digestive enzymes (cellulases, chitinases, proteases) and mucus.
* Passage through the gut increases available nutrient concentrations (available N, P, K) by a factor of 3 to 10 compared to raw waste.
- Vermicast Chemical Properties:
Vermicompost excretions (vermicasts) are granulated, water-stable micro-aggregates rich in microbial plant-growth promoting hormones (auxins, gibberellins, cytokinins) and humic acids, exhibiting superior CEC and moisture-holding properties compared to thermal compost.

>> Solution to Question 25: Analyze thermochemical Pyrolysis and Gasification of Solid Wastes, comparing syngas production with direct incineration and plasma

Fischer-Tropsch synthesis into liquid synthetic fuels.

3. Plasma Arc Gasification:

- Uses an electric plasma torch operating between tungsten electrodes at colossal temperatures: 3,000 to 10,000 deg C.
- The extreme thermal kinetic flux instantly disassociates hazardous chemical waste molecules directly into their constituent elemental atoms.
- Inorganic ash and heavy metals melt into an inert, non-leachable, vitrified glassy basaltic slag completely impervious to leaching, while organic matter converts into pure, tar-free synthesis gas.

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1. Statutory Regulatory Scope:
Enacted under the Environment (Protection) Act, 1986, the Bio-Medical Waste Management Rules apply to all healthcare facilities, veterinary hospitals, and medical research laboratories. A strict color-coded source-segregation architecture is legally mandated:
2. The Fourfold Color-Coded Segregation Architecture:
- 1. Yellow Color-Coded Non-Chlorinated Bags / Containers:
* Waste Stream: Human anatomical tissues, organs, amputated body parts, animal carcasses, soiled plaster casts, cotton gauze contaminated with blood/body fluids, expired cytotoxic pharmaceuticals, and microbiology cultures.
* Treatment & Disposal: Incineration in double-chamber incinerators (primary chamber >= 800 deg C, secondary >= 1050 deg C), or deep burial in remote areas.
- 2. Red Color-Coded Containers:
* Waste Stream: Contaminated recyclable plastics: intravenous tubing, catheters, urine bags, disposable plastic syringes (without needles), vacutainer tubes.
* Treatment: Thermal disinfection via Autoclaving or Hydroclaving, followed by mandatory mechanical shredding to mutilate items, and recycling through authorized recyclers.
- 3. White Translucent, Puncture-Proof, Tamper-Proof Containers:
* Waste Stream: Waste Sharps (used hypodermic needles, needles with fixed syringes, scalpels, surgical blades, contaminated glass slides).
* Treatment: Autoclaving or Dry Heat Sterilization, followed by encapsulation in concrete-filled steel drums or melting.
- 4. Blue Cardboard Boxes / Rigid Bins:
* Waste Stream: Broken or intact contaminated glassware (medicine vials, ampoules) and metallic orthopedic implants.
* Treatment: Disinfection by soaking in 1-2% Sodium Hypochlorite solution or autoclaving, followed by recycling.
3. Engineering Autoclaving Standards:
- Gravity flow autoclaves must operate at >= 121 deg C at 15 psi pressure for >= 60 minutes.
- Vacuum autoclaves must operate at >= 134 deg C at 31 psi for >= 45 minutes to achieve complete destruction of Geobacillus stearothermophilus endospores (spore log reduction >= 4).

>> **Solution Step Question 28** Explain hydrometallurgical and bioleaching metallurgical extraction processes for recovering Gold, Silver, and Copper from Electronic W

Gold is subsequently stripped and recovered via activated carbon adsorption or sodium borohydride chemical reduction.

3. Microbial Bioleaching (Bio-Hydrometallurgy):

- Employs extreme acidophilic chemolithoautotrophic bacteria (Acidithiobacillus ferrooxidans and Acidithiobacillus thiooxidans):

Microbes oxidize ferrous iron (Fe2+) and elemental sulfur, generating ferric sulfate [Fe2(SO4)3] and sulfuric acid in situ.

Ferric iron acts as an aggressive oxidant that chemically dissolves copper and nickel:

Cu0 (s) + 2 Fe3+ -> Cu2+ + 2 Fe2+.

- Cyanogenic heterotrophic bacteria (Pseudomonas putida, Chromobacterium violaceum) synthesize biogenic cyanide to leach gold as soluble dicyanoaurate [Au(CN)2]- under environmentally benign, mild conditions.

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recycles, and reclaims 100% of industrial wastewater, eliminating all liquid effluent discharge into environmental streams. All dissolved solids are precipitated out as dry, solid mineral salts suitable for commercial resale or secure landfilling.

2. The Multi-Stage ZLD Process Train:

- Stage 1: Physicochemical and Biological Pre-treatment:

High-rate Dissolved Air Flotation (DAF), Chemical Softening (Lime-Soda softening to precipitate Ca^{2+} and Mg^{2+} hardness down to < 10 mg/L), and Membrane Bioreactors (MBR) to remove suspended solids, oil/grease, and organic COD.

- Stage 2: Membrane Concentration (High-Pressure Reverse Osmosis, HPRO):

Effluent is concentrated across multiple RO stages (operating up to 80 to 120 bar hydrostatic pressure). Achieves 70% to 85% water recovery as ultra-pure permeate reused in factory boilers, leaving a concentrated brine containing 50,000 to 100,000 mg/L Total Dissolved Solids (TDS).

- Stage 3: Thermal Brine Concentrator with Mechanical Vapor Recompression (MVR):

* The RO reject brine enters a falling-film vertical tube evaporator.

* Thermodynamic MVR Principle: Water vapor boiling off the falling film at 100 deg C is compressed by a high-efficiency centrifugal mechanical compressor. Mechanical compression elevates vapor pressure and saturation temperature to 105-110 deg C.

* This superheated steam is re-injected into the shell side of the same evaporator tubes, condensing and transferring its latent heat of condensation to evaporate incoming brine! This eliminates continuous steam supply, slashing thermal operating energy by 80%. Concentrates brine up to 250,000 mg/L TDS (near saturation).

- Stage 4: Forced-Circulation Crystallizer and Centrifuge:

The saturated brine is pumped through a high-velocity heat exchanger into a crystallizer vessel where pure mineral crystals (NaCl, Na_2SO_4) nucleate and grow. The salt slurry is dewatered in solid-bowl centrifuges, yielding dry salt crystals for industrial reuse, closing the liquid loop completely.
